Original Article

A Comparison of the Therapeutic Effect of Euphorbia Macrolada Latex with Cryotherapy in the Treatment of Plantar Warts

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ABSTRACT

Skin warts are benign tumors caused by infection of keratinocytes with HPV and are visible as hyperkeratotic bumps. The aim was to compare the therapeutic effect of Euphorbia macrolada latex with cryotherapy in the treatment of plantar warts. In this study, 30 patients were examined and devided in two equal groups randomly. One group of patients was treated with latex of Euphorbia macrolada and the other group was treated with cryotherapy. Each patient was visited 4 times (0, 2, 4, and 12 weeks after the first treatment). Images of the lesions were captured and stored at a fixed, uniform distance by a single camera. In the group treated with the latex of euphorbia macrolada, the average wart size with 55.7% decreasing rate reached from 69.9 mm before treatment to 31 mm at the third visit (p=0.001) and two months after the last visit, it was decreased by 80% and reached to 13.9 mm (p=0.001) which in the second visit (two weeks later), all clinical findings were present in the cryotherapy group, but there was no pain or blistering in the euphorbia macrolada latex group and the incidence of tenderness, pain, and blistering in the cryotherapy group was significantly higher than other group. Considering the fewer complications and fewer recurrences of treating patients with the latex of Euphorbia macrolada, it seems that this treatment can be a suitable alternative for routine treatment with cryotherapy in the treatment of plantar warts.

Introduction

Skin warts are benign skin tumors caused by infection of keratinocytes with HPV [1]. Benign tumors of the skin and other epithelial tissues caused by infection with HPV, a small l double stranded DNA virus are generally called warts or papillomas [2]. Warts may have different forms depending on the epithelial surface and the type of HPV responsible. Common warts, plantar warts, genital warts, etc. are clinical manifestations of HPV infection [3].

Skin warts are common worldwide and are among the most common problems encountered by dermatologists and are also a common skin infection in children [4-8]. Skin warts can occur at any age, with a peak incidence in adolescence and early adulthood. Warts can occur anywhere on the skin [1], which can be physically and

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emotionally distressing [7], and are socially unacceptable when present in visible and exposed areas (such as the hands and face) [3]. They are a significant cause of anxiety and embarrassment in some patients and may interfere with social activities. The lesions may also be uncomfortable or bleed [10]. Transmission occurs through casual contact, especially at the site of skin damage and local spread is mostly due to reinoculation by the individual [1].

Indications for treatment include pain, interference with activities, social embarrassment, and risk of malignancy [10], and also the main goals of treatment are to eradicate the lesions, prevent the formation of ulcers and scars, and induce lifelong immunity to prevent relapse [8,10], while minimizing pain [8].

The lesions are notoriously difficult to treat and sometimes resolve spontaneously over time [9]. Spontaneous resolution of warts may take months to years, and it is impossible to provide a reliable prognosis for individual patients. However, it usually occurs more rapidly in children than in adults [1].

No single treatment is available that is effective in completely curing the patient, and as a result, there are many varied approaches to treating warts; for example, salicylic acid, cryotherapy, cantharidin, surgery, laser phototherapy, etc. [3,8].

As mentioned, one treatment method is cryotherapy. Cryotherapy with liquid nitrogen is widely used in the treatment of warts on the hands and feet [7,10,11] and involves freezing the wart with liquid nitrogen for 10 to 20 seconds, every 2 to 3 weeks [8].

The evidence for cryotherapy is conflicting [12-17]. The response to this treatment depends on many factors related to the patient and the wart [7].

Cryotherapy is quick and simple but is poorly tolerated by younger patients [8] and is in fact very expensive [18] and some authors believe that cryotherapy should not be used for more than 3 months or more than 4 times and there is no evidence that it can be continued for long periods, especially for plantar warts [19]. The main disadvantage of cryotherapy is that it can be painful and blistering and there is also a risk of permanent scarring, hyperpigmentation, and hypopigmentation [18]. Other side effects include blistering, skin irritation, and scaling [11].

Overall, treatment is complex, especially in children and repeated treatment of warts is tiring for doctors, patients, and caregivers. There is no complete cure or treatment that is quick, painless, and highly effective in reducing the risk of recurrence [8].

The knowledge of the medicinal properties of plants may be beyond the memory of history. The use of medicinal plants instead of synthetic drugs has increased in recent years due to the low incidence of side effects and the diversity of active plant compounds [20], and there is a diverse and extensive collection of medicinal plant species and the knowledge of their medicinal use as raw materials for the development of new drugs [21].

Euphorbiaceae is the largest family among the Anthophyta with 300 genera and 5,000 species [22]. This family contains a large variety of phytotoxins, mainly including esters, alkaloids, glycosides, and resin type toxins [23,24]. The type of Euphorbia is the largest type in the Euphorbiaceae with more than 200 species. All of them have latex and a unique flower structure [25]. Euphorbia species are used for treatment of various ailments, including skin diseases, gonorrhea, migraine, intestinal parasites, and warts [22]. Euphorbia has been used for hundreds of years in traditional Chinese medicine to treat warts [26]. In traditional Turkish medicine, Euphorbia species have been used for treament of rheumatism and edema, and especially to remove warts [23]. The species chosen for the study is Euphorbia helioscopia, which is native to temperate Eurasia, but has also adapted to subtropical conditions [24]. Numerous biological activities have been reported for this species in various sources, including vasopressor, antiallergic, antiasthmatic, insulin-secreting, antibacterial, antiviral, antifungal, antitumor, antioxidant, etc. [24].

In Iran, latex of Euphorbia helioscopia is also used topically for warts' treatment, and the mechanism of action has not been clearly mentioned in the sources studied [27,28].

The aim of this study was to compare the therapeutic effect of Euphorbia macrolada latex with cryotherapy in the treatment of plantar warts .

Materials and Methods

This study was a clinical trial. This trial was registered with the registration number IRCT20180807040734N1 in the Iranian Clinical Trials Registration Center at www.irct.ir. 60 patients with bilateral warts, regardless of gender, were included in the study after completing the consent form firstly. Of them, 4 patients were excluded from the study due to their unwillingness to cooperate, and 26 patients were excluded from the study due to their remote location, pain from cryotherapy, and poor patient cooperation. Finally, 30 patients were enrolled in this study and divided in two groups with 15 cases each. Patient information was recorded in pre-designed forms, including name and surname, age, gender, place of residence, marital status, etc. by the patients themselves and maintained by the administrators.

At the first visit, the size of the lesions on both sides was accurately measured with a caliper and recorded on special forms for each patient. Afterward, one group of patients was treated with latex of Euphorbia macrolada and the other group was treated with cryotherapy. Each patient was visited 4 times (0, 2, 4, and 12 weeks after the first treatment). Images of the lesions were taken and stored at a fixed interval by a single camera. This was repeated at each visit and before repeating the treatment. All information, including quantitative and statistical information as well as qualitative and visual information, was stored and kept until the end of the study, and at the end, the statistical information was entered into the software.

The image information obtained in each session and related to each patient was recorded with the date and compared at the end of the study.

Finally, collected data were entered into SPSS version 21, and then analyzed by table and graph and average and standard deviation. Ch-square and t-test were used to analyze differences and relationship between data.

Patients diagnosed with bilateral plantar warts who were willing to participate in the study and did not meet the exclusion criteria were selected. Patients with any systemic disease, pregnant women, and those with any side effects were excluded from the study.

Written informed consent was obtained from all patients to participate in the study. The code of the ethics committee for this study is IR.ARUMS.REC.1396.83.

Results

In this study, 30 samples were studied in two groups related to cryotherapy and latex of Euphorbia macrolada, each with 15 samples. Of the total patients, 11 (36.7%) were male, 17 (56.7%) were unemployed, and 17 (56.7%) were from urban population. The average age of all patients was 21.1± 5.9 (Table 1).

At the beginning of the study, there was no significant difference in the size of warts between the two groups, but in the second and the third visits, the difference between the two groups was statistically significant (Table 2).

Table 1: The demographic characteristics of patients in two groups

Variables		Cryotherapy		Euphorbia macrolada latex		*P*-value
		n	%	n	%	
Gender	Male	5	33.3	6	40	0.56
	Female	10	66.7	9	60	
Residence place	Urban	9	60	8	53.3	
	Rural	6	40	5	33.3	
Employee	Unemployed	8	53.3	9	60	
	Employed	5	33.3	6	40	

Table 2: Descriptive indices related to the size of warts at each visit

Appointment	Group therapy	Average wart size in mm	Group therapy	*P*-value
First visit	Cryotherapy	75	24.8	0.4
	Euphorbia macrolada latex	69.9	26.1	
Second visit	Cryotherapy	33.8	17.2	0.001*
	Euphorbia macrolada latex	49.8	18.8	
Third visit	Cryotherapy	6.1	9.6	0.001*
	Euphorbia macrolada latex	31	16.4	
2 months after the last visit	Cryotherapy	15.1	20.3	0.8
	Euphorbia macrolada latex	13.9	10.7	

*t-test

The average size of warts in the cryotherapy group decreased from 75 mm before treatment to 6.1 mm at the third visit, and 2 months after the last visit, the average size increased again and reached 15.1 mm, which was a 91.8% reduction in the third visit compared to the first visit and an 80% reduction two months after the last visit compared to the baseline. In the group related to the latex, the average size of warts decreased from 69.9 mm before treatment with a 55.7% reduction to 31 mm at the third visit (p=0.001) and the 80% reduction to 13.9 mm two months after the last visit (p=0.001) (Figure 1).

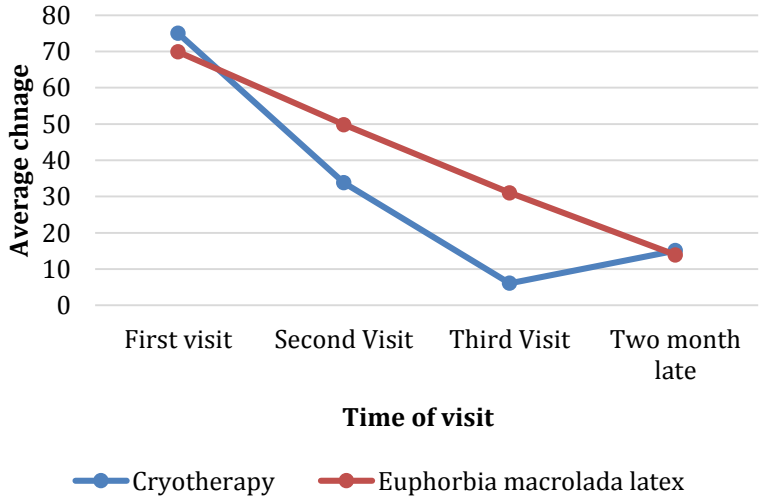


Figure 1: Average change in wart size in the two treatment groups during visits

At the second visit (two weeks later), all clinical findings were present in the cryotherapy group, but there was no pain or blistering in the macrolada latex group, and the incidence of tenderness, pain, and blistering in the cryotherapy group was significantly higher than in the macrolada latex group. Recurrence in the cryotherapy group was 147.5% and in the latex group was 55.7% (Table 3).

Table 3: Clinical findings (incidence of complications) in the two treatment groups by visit schedule

		Cryotherapy		Euphorbia macrolada latex		*P*-value
		n	%	n	%	
Second visit (2 weeks after the first treatment session)	Erythema	13	86.7	2	13.3	0.1
	Tenderness	9	60	0	0	0.001*
	Pain	11	70	0	0	0.001*
	Crust	9	63	12	80	0.4
	Blister	4	23.3	2	16.7	0.005*
Third visit (4 weeks after the first treatment session)	Erythema	8	53.3	1	6.7	0.003*
	Tenderness	8	53.3	2	13.3	0.038*
	Pain	5	33.3	3	20	0.02*
	Crust	15	100	6	40	0.015*
	Blister	0	0	0	0	0
2 months after the last visit (12 weeks after The first treatment session)	Erythema	0	0	0	0	0
	Tenderness	2	13.3	0	0	0.034*
	Pain	1	6.7	0	0	0.3
	Crust	0	0	0	0	0
	Blister	0	0	0	0	0

*chi-square test

Discussion

This clinical trial study has compared the therapeutic effect of using latex of Euphorbia helioscopia with cryotherapy in the treatment of plantar warts. To use these two treatment methods, the groups under study should be in approximately the same range in terms of age and gender; considering that both treatment methods used on patients were performed on individuals simultaneously, the latter case has been well observed.

Several treatment modalities for warts have been investigated and studied in recent years, but none of them have been associated with 100% cure. However, cryotherapy with liquid nitrogen is still the most popular treatment for warts in patients due to its easy of use and effectiveness, as this treatment method can cure 50-70% of the patient's lesions after 3 to 4 treatment sessions. In a study by Shahab Bohlooli *et al.*, that investigated the effect of fig juice extract in the treatment of warts compared to cryotherapy, it was concluded that fig extract has a shorter duration than cryotherapy [25]. Due to the high pain and high recurrence rate in cryotherapy treatment, studies have always been conducted to replace this method. On the other hand, considering the mechanism of wart formation through benign proliferation of skin cells due to the involvement of the virus in the cell cycle of the involved cells, it seems that neutralizing the effect of the virus in the relevant cells can prevent the formation of new warts or the gradual treatment of old lesions. In a study by Ardabili *et al.*, on the effect of cimetidine on the treatment of warts, it was shown that cimetidine was not more effective than placebo in the treatment of warts [26]. One possible therapeutic modality in controlling cell proliferation is the use of Euphorbia plant extract. Recently, a study has shown that the latex of the plant modulates gene expression at the site of laryngeal SCC lesions and thus significantly reduces cell growth at the site of the lesion [1]. Although some studies have shown that Euphorbia extract is effective in treating warts, they have not necessarily evaluated this method as more effective than other methods. For example, in a study conducted by Yazdanfar *et al.*, no significant difference was found

between the standard treatment method (composition of salicylic acid 16.7 g and lactic acid 16.7 g in 6.66 g of collodion) and the use of Euphorbia macroclada extract [27]. Also, the use of the Euphorbia helioscopia in traditional medicine of various countries such as Turkey, China, and Iran in the treatment of skin lesions such as warts can itself suggest the hypothesis of the positive effect of this plant in the treatment of these lesions. In the present study, although the short-term therapeutic effect of cryotherapy was significantly greater than the use of Euphorbia helioscopia extract, in the long-term study, the therapeutic effect of both treatment methods was estimated to be the same; however, the short-term complications of using the less invasive method of latex were significantly lower than the other method. Sayadrezaei *et al*. compared the therapeutic effect of topical fluorouracil with cryotherapy in the treatment of warts. In this study, it was concluded that although fluorouracil treatment has a longer treatment duration and a similar recurrence rate to cryotherapy, its advantage is that it is painless [28]. In the study of Shahab Bohlooli *et al.*, cryotherapy treatments, like the present study, were preferable to cryotherapy in terms of being painless and having fewer complications, including tenderness, pain, blistering, and crusting. The limitations of the present study include the small number of patients, withdrawal of patients from the study due to pain from cryotherapy, remoteness of the place of residence, and poor cooperation of the patient, seasonal restrictions related to the plant, the absence of extensive studies related to the plant and the leachate, and lack of knowledge about how to maintain and care for the treatment site [25].

Conclusion

Considering the fewer complications and recurrences of patients treated with latex of Euphorbia helioscopia, it seems that this method can be a suitable alternative for routine cryotherapy treatment in the treatment of plantar warts. It is recommended that a study with a larger sample size and longer follow-up periods be conducted to investigate the long-term complications of EH treatment. It is also better to conduct a study to compare the therapeutic results of EH latex with other therapeutic methods to select the best treatment method.

Conflict of Interest

None.

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